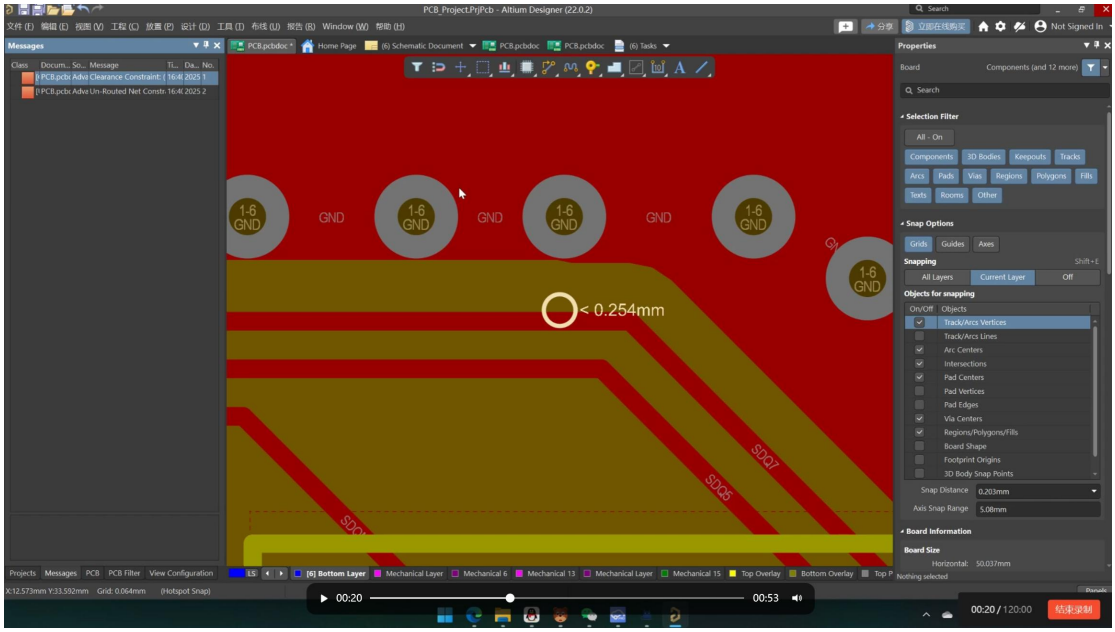
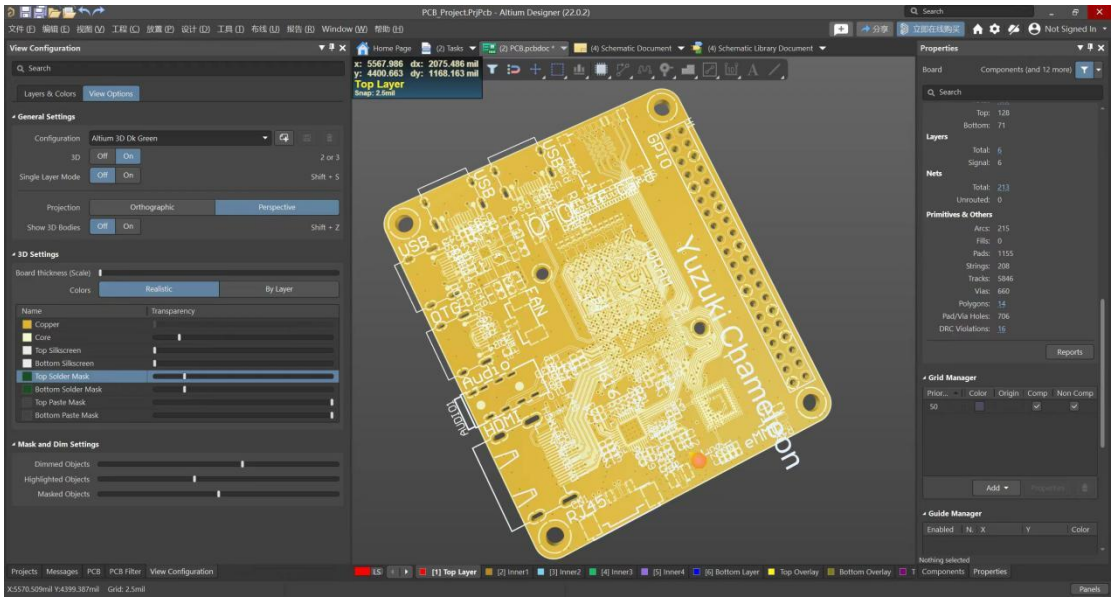
H616 调试笔记

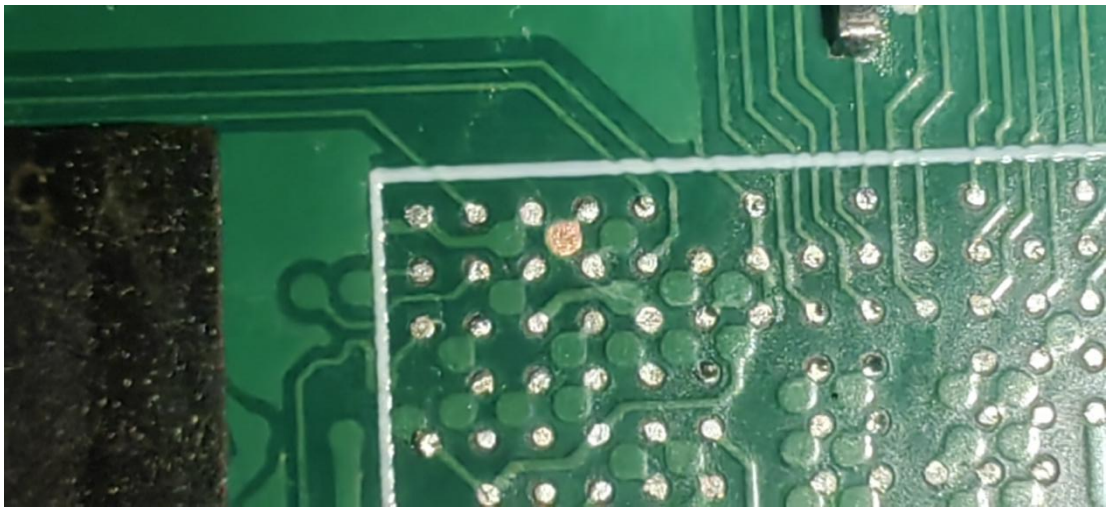
1.使用别人的嘉立创开源工程导入 AD，发现预览异常，且导出焊接辅助工具预览时异常，且出现奇怪报错





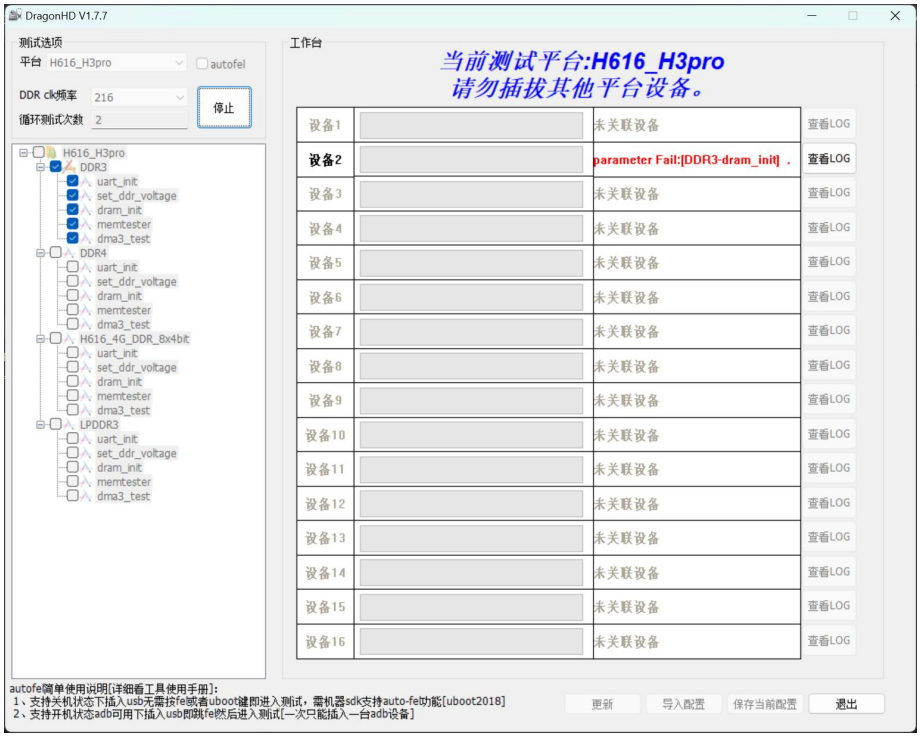
原因：不同电路板软件之间有兼容问题，不要中途改变设计软件

2.第一次焊接的时候把主芯片焊反了，PCB 出现阻焊脱落，且在通用植锡网里找不到适配的

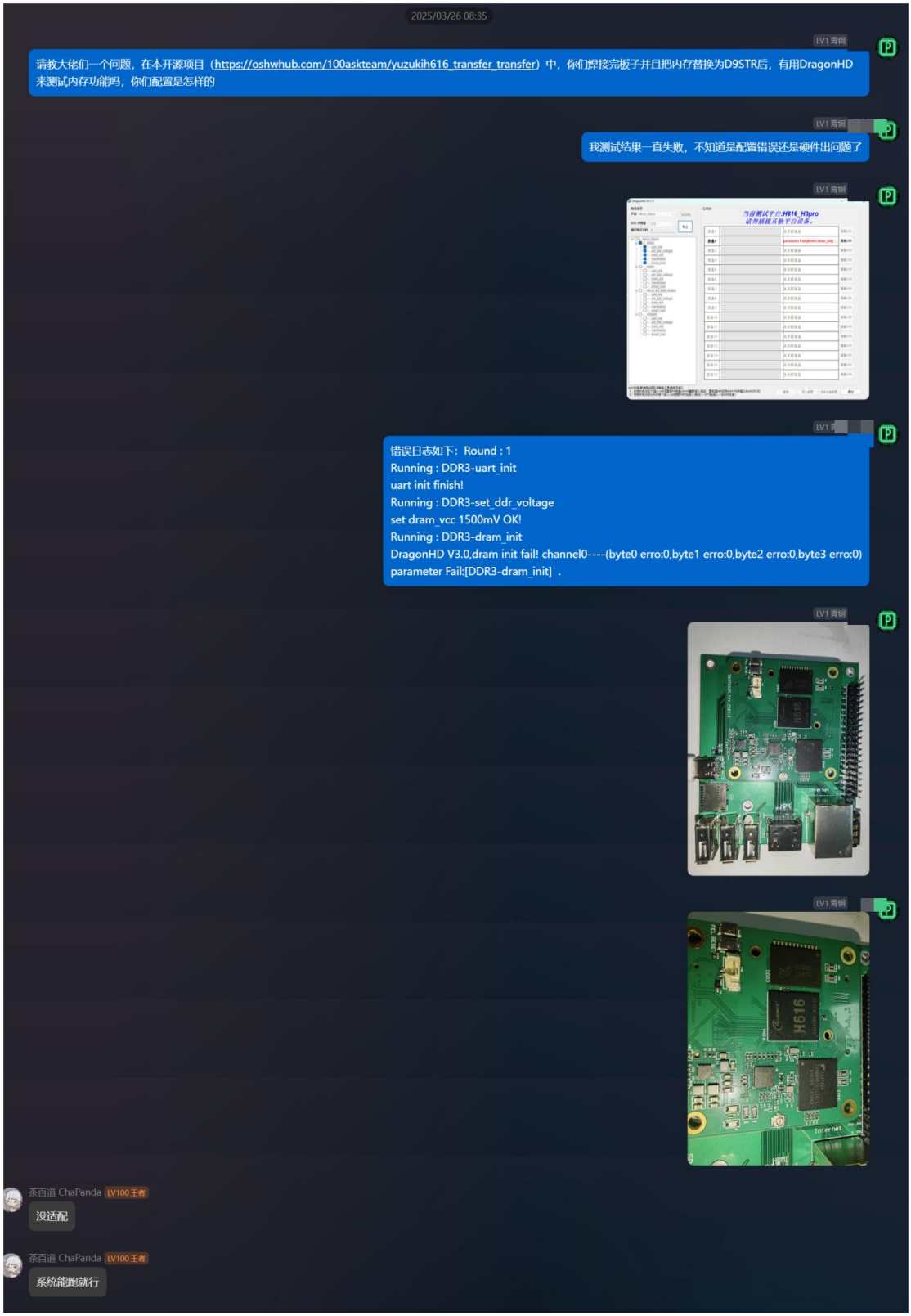


解决方法：要购买专用的植锡网，需要使用绿油修补，然后紫光灯固化阻焊，防止短路

3.焊接完板子并且把内存替换为 D9STR 后，用 DragonHD 来测试内存功能，测试结果一直失败



原因：测试平台没有对此款内存进行适配



4.一直找不到合适的镜像

LV1 青

请教大佬们一个问题，我焊好本开源工程（https://github.com/YuzukiHD/YuzukiChameleon/tree/master?tab=readme-ov-file）的板子后，想刷Ubuntu图形化操作系统镜像来测试板子功能，但是我刷入仓库中release里面的H616_YuzukiChameleon_3.0.0_debian_bullseye_server_linux5.16.17.img.7z后，发现板子启动后HDMI只出现Linux命令行。我后面又寻找了一遍仓库，包括仓库里提到的另一个主线Linux仓库，都没找到已经编译好的图形化操作系统镜像，最多只有源码，请问图形化操作系统镜像在哪里

2025/03/26 12:24

AAAA土方龙哥 LV23 黄金

> 请教大佬们一个问题，我焊好本开源工程（https://github.com/YuzukiHD/YuzukiChameleon/tree/master?tab=readme-ov-file）的板子后，想刷Ubuntu图形化操作系统镜像来测试板子功能，...

server代表服务器版本

AAAA土方龙哥 LV23 黄金

请使用avaotaos自行编译

2025/03/26 16:01

LV1 青铜

> AAAA土方龙哥
> 请使用avaotaos自行编译

请教大佬们一个问题，我焊好板子后，想刷Ubuntu图形化操作系统镜像来测试板子功能，群友说使用avaotaos编译，我在网上找到了其使用文档"https://docs.avaota.fun/linux-manual/avaotaos/os-flash"，却发现其开头写使用Avaota Pi开发板，可是这个开发板使用的主控不是全志H616，不知道能不能照着这个教程尝试

黑糖珍珠 LV100 管理员

通用的

黑糖珍珠 LV100 管理员

616和618

Quad-core ARM Cortex-A76 and Quad-core Cortex-A55 LV100 王者

chameleon有avaotaos了喵?

Quad-core ARM Cortex-A76 and Quad-core Cortex-A55 LV100 王者

可以把我吃灰的chameleon拿出来用了喵

噬血狂袭 LV30 黄金

哪个群友说的能用你去找哪个群友

噬血狂袭 LV30 黄金

让他给一下支持

2025/03/26 16:53

LV1 青铜

> 黑糖珍珠
> 616和618

@黑糖珍珠 我刚刚照着avaotaos的教程"https://docs.avaota.fun/linux-manual/avaotaos/os-flash"的指引，下载了"https://github.com/AvaotaSBC/AvaotaOS/releases"的桌面版本"AvaotaOS-0.3.0.4-noble-gnome-arm64-avaota-a1.img.xz"，烧录后发现板子是黑屏的，不知道是怎么回事（之前已经烧录了"https://github.com/YuzukiHD/YuzukiChameleon/tree/master?tab=readme-ov-file"的release的镜像能跑，说明板子是没问题的）

黑糖珍珠 LV100 管理员

看串口

2025/03/26 17:11

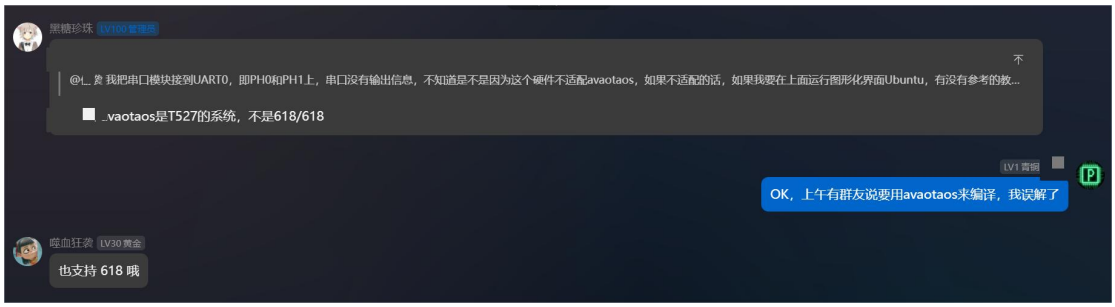
LV1 青铜

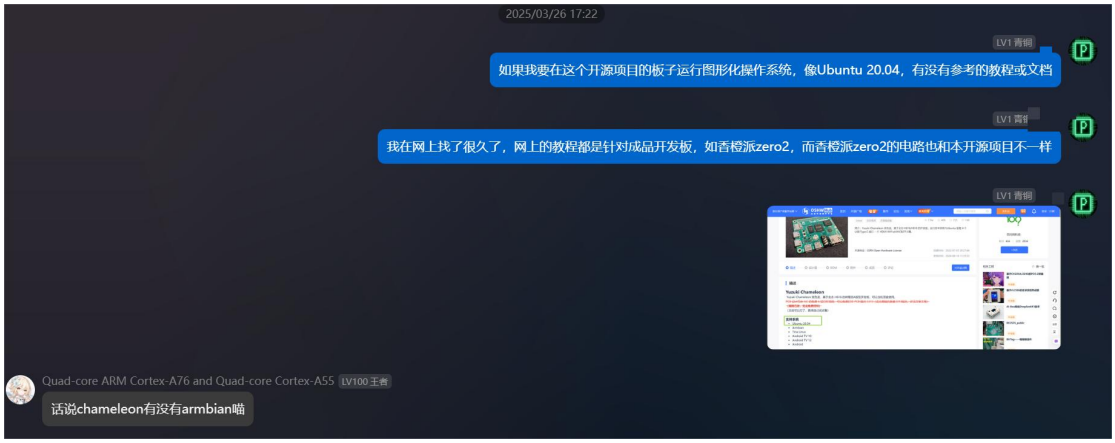
> 黑糖珍珠
> 看串口

@黑糖珍珠 我把串口模块接到UART0，即PH0和PH1上，串口没有输出信息，不知道是不是因为这个硬件不适配avaotaos，如果不适配的话，如果我要在上面运行图形化界面Ubuntu，有没有参考的教程或文档

噬血狂袭 LV30 黄金

给你说了 AvaotaOS 不支持 Chameleon





原因：这个开源工程相关资料并没有消费级开发板那么完善，最好直接复刻消费级开发板方案

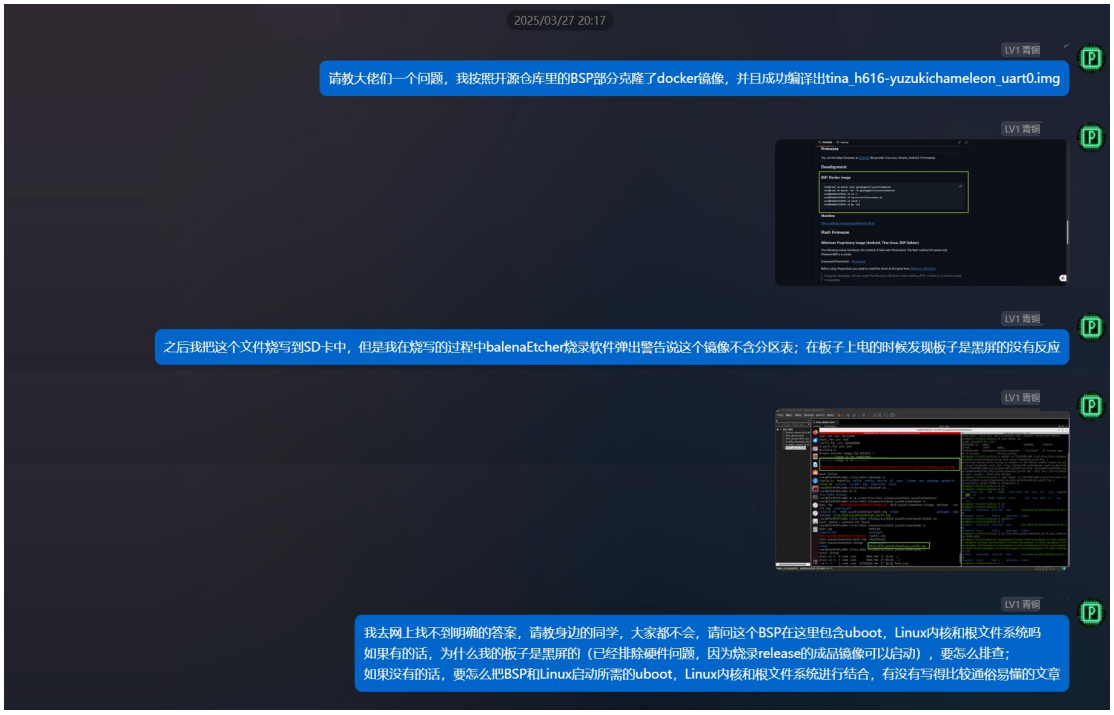
5. docker 镜像一直无法下载

docker pull 报错Get “https://registry-1.docker.io/v2/ “: net/http: request canceled while waiting for c

解决办法：切换网络

6.烧录 docker 编译的镜像后，黑屏

原因：emmc 和 SD 卡镜像烧录方式不一样，且编译配置不同

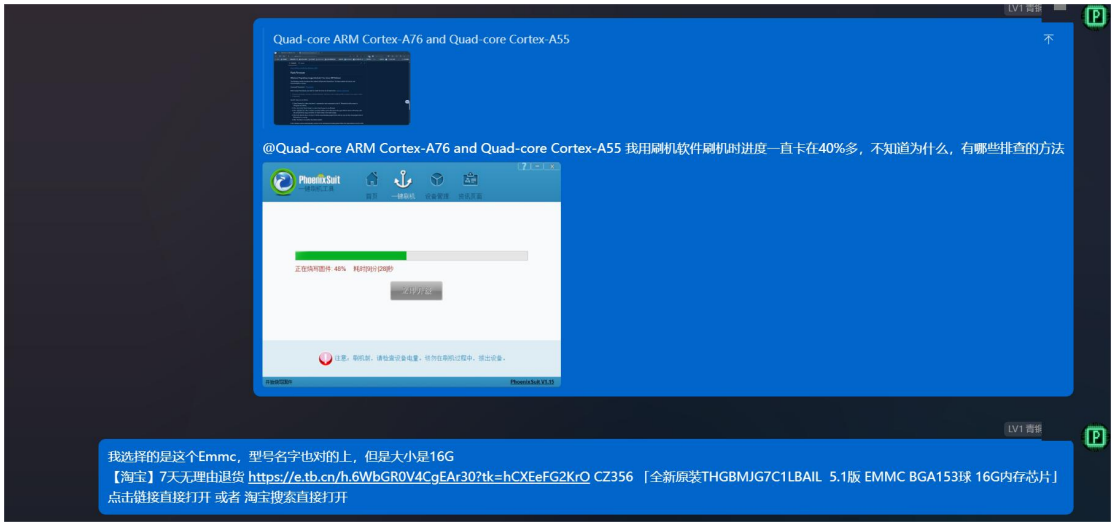


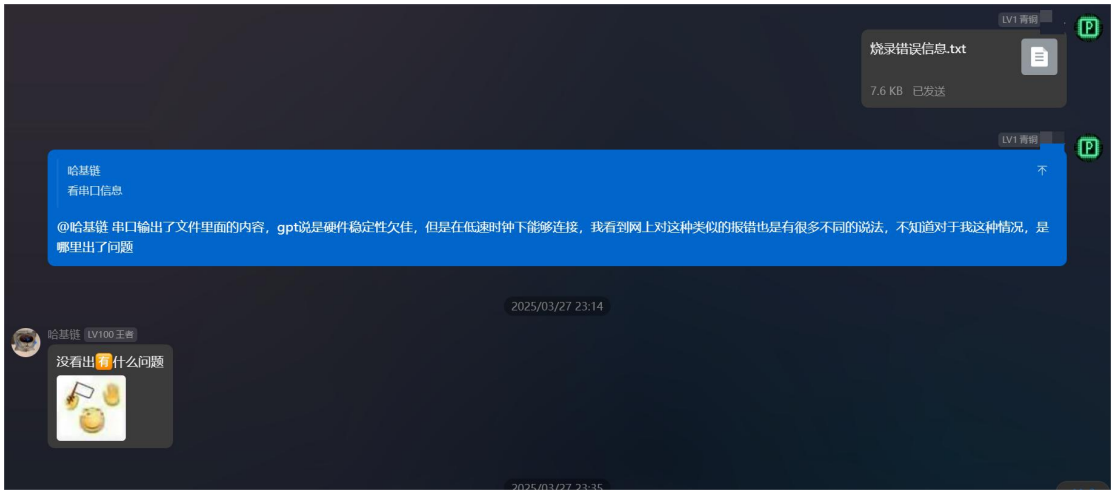
7.EMMC 镜像无法烧录，进度条卡住，

错误信息：

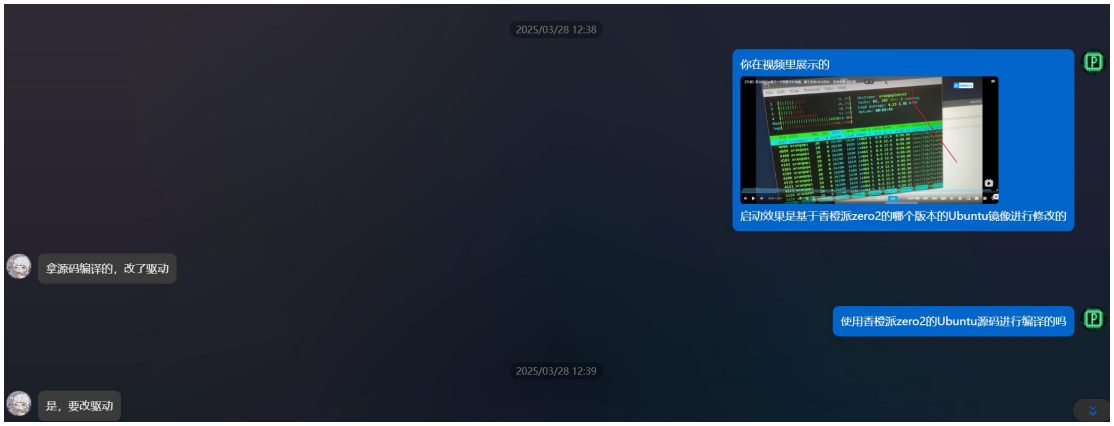
```
[20335]fes begin commit:a00ebdcb84
[20338]set pll start
[20341]periph0 has been enabled
[20344]set pll end
[20347]unknow PMU
[20349]unknow PMU
[20353]PMU: AXP1530
[20358]vaild para:1    select dram para0
[20361]board init ok
[20363]beign to init dram
[20366]DRAM BOOT DRIVE INFO: V0.651
[20369]the chip id is 0x5000
[20372]chip id check OK
[20394]DRAM_VCC set to 1500 mv
[20400]read_calibration error
[20406]read_calibration error
[20412]read_calibration error
[20418]read_calibration error
[20423]read_calibration error
[20429]read_calibration error
[20435]read_calibration error
[20441]read_calibration error
[20447]read_calibration error
[20453]read_calibration error
[20456]retraining final error
[20461]read_calibration error
[20467]read_calibration error
[20473]read_calibration error
[20479]read_calibration error
[20484]read_calibration error
[20490]read_calibration error
[20496]read_calibration error
[20502]read_calibration error
[20507]read_calibration error
[20513]read_calibration error
[20516]retraining final error
[20522][AUTO DEBUG]16 bit,2 ranks training success!
[20533]DRAM CLK =792 MHZ
[20535]DRAM Type =3 (3:DDR3,4:DDR4,7:LPDDR3,8:LPDDR4)
[20544]Actual DRAM SIZE =2048 M
[20547]DRAM SIZE =2048 MBytes, para1 = 310a, para2 = 8001001, dram_tpr13 = 6041
[20556]DRAM simple test OK.
[20558]rtc standby flag is 0x0, super standby flag is 0x0
[20564]init dram ok


U-Boot 2018.05 (Mar 27 2025 - 09:22:23 +0000) Allwinner Technology

[23.723]CPU:      Allwinner Family
[23.726]Model: sun50iw9
I2C:      ready
[23.730]DRAM:    512 MiB
[23.733]Relocation Offset is: 15ebf000
[23.768]secure enable bit: 0
[23.772]pmu_axp152_probe pmic_bus_read fail
[23.776]PMU: AXP1530
[23.777]PMU: pmu_axp1530 found
[23.783]dcdc1_vol = 960, onoff=1
[23.786]dcdc2_vol = 1000, onoff=1
[23.790]aldo1_vol = 1800, onoff=1
[23.793]dldo1_vol = 3300, onoff=1
[23.796]CPU=1008 MHz,PLL6=600 Mhz,AHB=200 Mhz, APB1=100Mhz    MBus=400Mhz
[23.802]gic: normal mode
[23.805]try fast burn key
[23.808]out of usb burn from boot: not boot mode
[23.813]flash init start
[23.815]workmode = 16,storage type = 0
try card 2
set card number 2
get card number 2
[23.823][mmc]: mmc driver ver uboot2018:2021-07-19 14:09:00
[23.829][mmc]: get sdc_type fail and use default host:tm4.
[23.840][mmc]: Is not Boot mode!
[23.843][mmc]: SUNXI SDMMC Controller Version:0x40502
[23.854][mmc]: ************Try SD card 2************
[23.859][mmc]: mmc 2 cmd timeout 100 status 100
[23.863][mmc]: smc 2 err, cmd 8,    RTO
[23.867][mmc]: mmc 2 close bus gating and reset
[23.872][mmc]: mmc 2 cmd timeout 100 status 100
[23.876][mmc]: smc 2 err, cmd 55,    RTO
[23.879][mmc]: mmc 2 close bus gating and reset
[23.884][mmc]: ************Try MMC card 2************
[23.926][mmc]: mmc 2 cmd timeout 100 status 100
[23.930][mmc]: smc 2 err, cmd 8,    RTO
[23.933][mmc]: mmc 2 close bus gating and reset
[23.938][mmc]: mmc 2 cmd timeout 100 status 100
[23.942][mmc]: smc 2 err, cmd 55,    RTO
[23.946][mmc]: mmc 2 close bus gating and reset
[23.980][mmc]: gen_tuning_blk_bus8: total blk 10
[23.984][mmc]: gen_tuning_blk_bus4: total blk 6
[23.988][mmc]: Using 8 bit tuning now
[23.993][mmc]: write_tuning_try_freq: write ok
[23.998][mmc]: Pattern compare ok
[24.001][mmc]: Write tuning pattern ok
[24.005][mmc]: ================== HSSDR52_SDR25...
[24.010][mmc]: skip freq 400000
[24.012][mmc]: skip freq 25000000
[24.015][mmc]: freq: 2-50000000-64-4
[24.202][mmc]: [0-40|41] [53-63|11]
[24.205][mmc]: ================== HSDDR52_DDR50...
[24.210][mmc]: skip freq 400000
[24.213][mmc]: freq: 1-25000000-64-4
[24.402][mmc]: freq: 2-50000000-64-4
[24.551][mmc]: [0-34|35] [42-42|1] [46-55|10] [58-63|6]
[24.556][mmc]: [0-28|29] [30-34|5] [43-63|21]
[24.561][mmc]: DS26/SDR12: 0xffffffff 0xffffffff
[24.565][mmc]: HSSDR52/SDR25: 0xff14ffff 0xffffffff
[24.570][mmc]: HSDDR52/DDR50: 0xff0e11ff 0xffffffff
[24.574][mmc]: HS200/SDR104: 0xffffffff 0xffffffff
[24.579][mmc]: HS400: 0xffffffff 0xffffffff
[24.583][mmc]: HS400: 0xffffffff 0xffffffff
[24.587][mmc]: Best spd md: 2-HSDDR52/DDR50, freq: 2-50000000, Bus width: 8
[24.595]Loading Environment from SUNXI_FLASH... OK
[24.599]Net:      [24.601]Net Initialization Skipped
[24.604]No ethernet found.
Hit any key to stop autoboot:    0
sunxi work mode=0x10
run usb efex
delay time 2500
weak:otg_phy_config
usb init ok
set address 0x20
set address 0x20 ok
set address 0x22
set address 0x22 ok
SUNXI_EFEX_ERASE_TAG
erase_flag = 0x0
FEX_CMD_fes_verify_status
FEX_CMD_fes_verify last err=0
the 0 mbr table is ok
the 1 mbr table is ok
the 2 mbr table is ok
the 3 mbr table is ok
*************MBR DUMP****************
total mbr part 8

part[0] name          :bootloader
part[0] classname :DISK
part[0] addrlo       :0x8000
part[0] lenlo        :0x10000
part[0] user_type :32768
part[0] keydata     :0
part[0] ro            :0

part[1] name          :env
part[1] classname :DISK
part[1] addrlo       :0x18000
part[1] lenlo        :0x8000
part[1] user_type :32768
part[1] keydata     :0
part[1] ro            :0

part[2] name          :boot
part[2] classname :DISK
part[2] addrlo       :0x20000
part[2] lenlo        :0x10000
part[2] user_type :32768
part[2] keydata     :0
part[2] ro            :0

part[3] name          :rootfs
part[3] classname :DISK
part[3] addrlo       :0x30000
part[3] lenlo        :0x893e0
part[3] user_type :32768
part[3] keydata     :0
part[3] ro            :0

part[4] name          :rootfs_data
part[4] classname :DISK
part[4] addrlo       :0xb93e0
part[4] lenlo        :0x8000
part[4] user_type :32768
part[4] keydata     :0
part[4] ro            :0

part[5] name          :misc
part[5] classname :DISK
part[5] addrlo       :0xc13e0
part[5] lenlo        :0x8000
part[5] user_type :32768
part[5] keydata     :0
part[5] ro            :0

part[6] name          :private
part[6] classname :DISK
part[6] addrlo       :0xc93e0
part[6] lenlo        :0x8000
part[6] user_type :32768
part[6] keydata     :0
part[6] ro            :0

part[7] name          :UDISK
part[7] classname :DISK
part[7] addrlo       :0xd13e0
part[7] lenlo        :0x0
part[7] user_type :33024
part[7] keydata     :0
part[7] ro            :0

total part: 9
mbr 0, 8000, 8000
bootloader 1, 10000, 8000
env 2, 8000, 8000
boot 3, 10000, 8000
rootfs 4, 893e0, 8000
rootfs_data 5, 8000, 8000
misc 6, 8000, 8000
private 7, 8000, 8000
UDISK 8, 0, 8100
[58.127]erase all part start
need erase flash: 0
******Has init
private part exist
GPT:bootloader    : 12000              21fff
GPT:env                : 22000              29fff
GPT:boot               : 2a000              39fff
GPT:rootfs            : 3a000              c33df
GPT:rootfs_data : c33e0              cb3df
GPT:misc               : cb3e0              d33df
GPT:private          : d33e0              db3df
GPT:UDISK              : db3e0              1d59fde
begin to store data
part 0 name bootloader
keydata = 0x0
part 1 name env
keydata = 0x0
part 2 name boot
keydata = 0x0
part 3 name rootfs
keydata = 0x0
part 4 name rootfs_data
keydata = 0x0
part 5 name misc
keydata = 0x0
part 6 name private
keydata = 0x0
find keypart private
keypart read start: 0xc93e0, sectors 0x8000
keypart part private read    0xc93e0, sectors 0x8000
part 7 name UDISK
keydata = 0x0
begin to erase
[58.410]erase all part start
finish erase
******Has init
rewrite
******Has init
write primary GPT success
write Backup GPT success
keypart write start: 0xc93e0, sectors 0x8000
flash exit
SUNXI_EFEX_MBR_TAG
mbr size = 0x10000
******Has init
write primary GPT success
write Backup GPT success
FEX_CMD_fes_verify_status
FEX_CMD_fes_verify last err=0
******Has init
FEX_CMD_fes_verify_value, start 0x8000, size high 0x0:low 0x14000
FEX_CMD_fes_verify_value 0xe6f75b5c
FEX_CMD_fes_verify_value, start 0x18000, size high 0x0:low 0x20000
FEX_CMD_fes_verify_value 0x166c4051
```

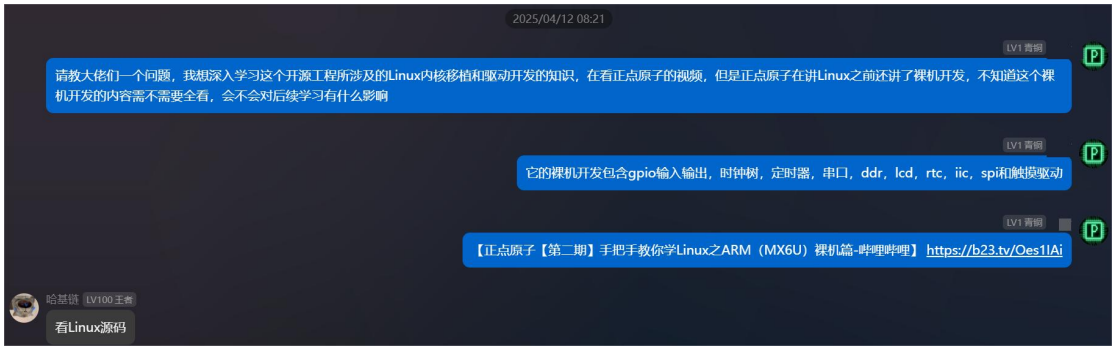
原因：尚未解决，可能是因为 EMMC 型号不对

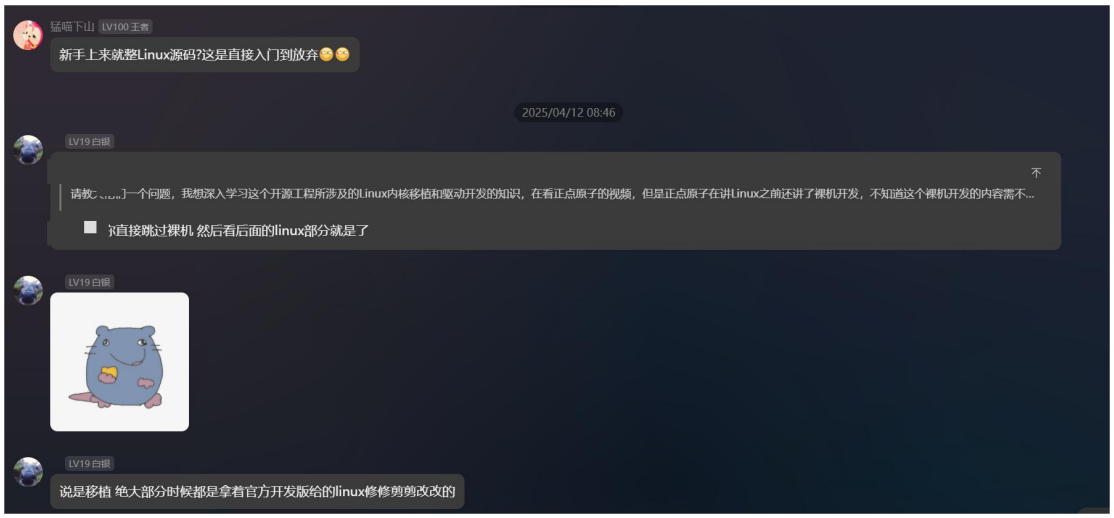




8.尝试复现视频里出现的香橙派系统，失败了







原因：还不会 Linux 内核和系统移植，和进阶驱动开发